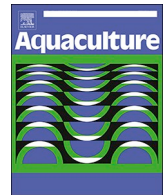




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Dietary menthol essential oil enhanced the growth performance, digestive enzyme activity, immune-related genes, and resistance against acute ammonia exposure in Nile tilapia (*Oreochromis niloticus*)

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ABSTRACT

Dietary supplementation of herbal essential oils is recommended for aquatic animals due to their growth-promoting, immune, and antioxidative potential. Hence, the present study evaluated the inclusion of menthol essential oil (MNT) at 0.1%, 0.2%, 0.3%, and 0.4% on the growth performance, digestive enzyme activity, immunity, and antioxidative responses of Nile tilapia. After eight weeks, samples were collected, and fish was exposed to acute ammonia challenge at 5 mg TA-N/L for 6 h. The results displayed that dietary MNT significantly enhanced the final body weight (FBW), weight gain, and feed conversion ratio (FCR) in a linear and quadratic manner ($P < 0.05$). The regression analysis revealed that the optimum rate of dietary MNT required for the best FBW is 0.26% and 0.30% for FCR. Dietary MNT enhanced the protease activity at the rate of 0.30% ($P < 0.05$) while lipase and amylase activities did not influence by MNT supplementation. Before and after the ammonia challenge, lysozyme (LZM), phagocytic activities (PA), phagocytic index (PI), superoxide dismutase (SOD), catalase (CAT), and glutathione peroxidase (GPx) were higher in fish fed MNT than fish fed MNT free diet. On the other hand, cortisol, glucose, and malondialdehyde (MDA) were decreased by dietary MNT in a linear and quadratic manner ($P < 0.05$). Ammonia stress induced reduced LZM, PA, and PI but increased the SOD, CAT, GPx, MDA, cortisol, and glucose levels. The related inflammatory genes (*IL-8*, *TNF- α* , and *IL-1 β*) were reduced by dietary MNT either before or after the ammonia exposure, and the mRNA levels of these genes were increased after ammonia stress. The ammonia exposure upregulated the expression of HSP70, and dietary MNT displayed reduced HSP70. The study assumed that MNT inclusion at the rate of 0.22% to 0.30% is recommended to increase the growth performance, protease activity, immunity, antioxidative response, and the resistance against ammonia exposure in Nile tilapia.

1. Introduction

Fish and its products are increasingly required for global food security, which is attributed to the rich nutritional value and reasonable cost per gram protein compared to other animal protein sources (Dawood, 2016; FAO, 2018). For that, the aquaculture is recommended as the most growing food sector based on the rearing of several valued aquatic animals (Garlock et al., 2020). Nile tilapia (*Oreochromis niloticus*) is among the main cultured fish species due to its high nutritional and commercial value as well as the tolerance against stressful conditions (Dawood et al., 2020a). Therefore, it can be reared under

intensive and super-intensive aquaculture systems. However, the intensive rearing conditions induce several stressful impacts that impair the growth rate and health status of fish (Dawood et al., 2019a). The high amounts of ammonia are the primary toxic pollutants that abundantly exists in the aquatic environment (Hoseini et al., 2019). The danger of ammonia accumulation in the rearing water-induced decreased growth rate, feed efficiency, and survivability of aquatic animals (Taheri Mirghaied et al., 2019). Additionally, it suppresses the immune system and diminishes the antioxidative status (Cheng et al., 2015; Qi et al., 2017). Based on that, the fish becomes more susceptible to infectious diseases and outbreaks. In such cases, antibiotics are used

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to treat the fish; however, the over and continuous application of chemotherapies causes high resistant of pathogenic bacteria, hazards to the ecosystem, and unfavourable impacts on human health (Akanmu, 2018). Therefore, looking for solid alternative substances is a crucial choice for sustainable and healthy aquaculture activity (Hoseini et al., 2020; Yilmaz, 2019). Under the conditions of ammonia toxicity, the application of medicinal herbs and its extracts has been proven to potentiate the growth performance and wellbeing of aquatic animals (Hoseini et al., 2019; Taheri Mirghaied et al., 2019).

Medicinal herbs are known to be beneficial, adequate, and eco-friendly alternatives to antibiotics (Dawood et al., 2018; Yilmaz and Ergün, 2018). Furthermore, medicinal herbs have useful natural derivatives that have several benefits, including improvement of the gastrointestinal microbiota, growth performance, feed utilization, and the improvement of the host's immune status (Dawood, 2020; Gabriel, 2019; Reverter et al., 2014). Herbal essential oils have been found to potentiate beneficial influences on the performances of aquatic animals (Abdel-Latif et al., 2020; Dawood, 2020). Its use has shown positive results in fish culture, including the enhancement of growth, survival, improvement of innate immune responses, and disease prevention (Abdel-Latif et al., 2020b). The inclusion of myrcene, menthol (Hoseini et al., 2019), and 1,8-cineole (Taheri Mirghaied et al., 2019) in the diets of common carp (*Cyprinus carpio*) improved the growth performance, antioxidative status, immunity, and resistance against ambient ammonia exposure. Further, Yilmaz (2020) illustrated that Nile tilapia fed diets enriched with carob (*Ceratonia siliqua*) syrup showed enhanced growth performance, immunity, and resistance against ammonia exposure. Recently, dietary menthol essential oil resulted in improving the growth rate, antioxidative, and anti-inflammatory responses in Nile tilapia exposed to chlorpyrifos toxicity (Dawood et al., 2020e). However, no studies were done to clarify the possibility of using dietary MNT in tilapia exposed to ammonia exposure. Therefore, the present study aimed to evaluate growth performance, digestive enzyme activity, antioxidative response, immune-related genes, and resistance against acute ammonia exposure in Nile tilapia fed dietary MNT.

2. Materials and methods

2.1. Experimental diets

The experimental diets were prepared by following Dawood et al. (2020e), where the basal diet was supplemented with four levels of menthol essential oil (MNT, 98% purity, Al-Hedaia Group CO., El-Gharbia, Egypt). The basal diet was assigned as the control group (0% MNT) and 4 supplemented diets with MNT at 0.1%, 0.2%, 0.3%, and 0.4% (Table 1). The ingredients of the formulated diets were mixed in the presence of water and oil to produce a dough by the laboratory meat grinder (1–2 mm). The produced pellets were air-dried and kept at 4 °C

Table 1
Basal diet and proximate chemical composition (on dry matter basis).

Ingredient	(%)	Chemical composition	(%)
Fish meal (65%)	10	Crude protein	30.14
Soybean meal (44%)	36	Crude lipids	6.98
Gluten	5	Ash	7.57
Wheat bran	12	Crude fibers	5.45
Rice bran	10	Gross energy (kcal/kg) ²	1844.51
Yellow corn	12		
Wheat flour	8.92		
Fish oil	3		
Vitamin and mineral mix ¹	2		
Dicalcium phosphate	1		
Vitamin C	0.08		

¹ Vitamin and mineral mixture detailed by Dawood et al. (2020b).

² Gross energy was calculated based on the values of values for protein, lipid, and carbohydrate as 23.6, 39.5 and 17.2 KJ/g, respectively.

until use. The formulated diets' chemical composition was confirmed by following the standard methods (AOAC, 2007).

2.2. Fish and rearing trial

Nile tilapia were collected from a local farm and transported gently to the laboratory for acclimatization. Fish were kept for two weeks in 500 L tanks with continuous aeration and fed the basal diet three times a day. After that, healthy and homogenous sizes of fish with an average initial weight (15.11 ± 0.06 g) were randomly allocated into 15 glass aquaria (5 groups/triplicates), which placed with 60 L of water (15 fish in each aquarium). Each aquarium was supplied with continuous aeration, and the water was changed daily with fresh dechlorinated water. The prepared diets were included in each tank at the rate of 3% two times a day (08:00 am and 03:00 pm) for eight weeks (12 h day: 12 h dark). The fish weighed every two weeks to recalculate the amount of feed intake and visually check the health condition of fish. During the trial, the physical water characteristics were checked regularly. The water temperature, pH, dissolved oxygen, and total ammonia values were 26.2 ± 0.7 °C, 7.6 ± 0.8 , 6.2 ± 0.4 mg/L, 0.18 ± 0.02 mg/L, respectively. Water temperature, dissolved oxygen, and pH were measured by mercury thermometer, Mettler Toledo (model 128.s/No1242), and pH meter (Extech pH/temp pen model pH 60), accordingly. Total ammonia was measured calorimetrically.

2.3. Final sampling

At the end of the feeding trial (8 weeks), all fish were deprived of feeding 24 h before final sampling. The fish were weighed and counted individually to calculate the weight gain, specific growth rate, feed conversion ratio (FCR), and survival rate (Table 3). Then, fish were anesthetized with tricaine methanesulphonate (MS-222) at 25 mg/L, and blood was collected from the caudal vein (3 fish/aquarium) using non-heparinized syringes. Collected blood was kept in Eppendorf tubes for a while, centrifuged at 3000 rpm/ 15 min at 4 °C, and the serum was collected and kept at -20 °C for further analysis. The hepatic tissue was dissected from the same fish and immediately frozen in liquid nitrogen and transported to -80 °C in the laboratory freezer. The intestines (midgut) were quickly removed and kept in liquid nitrogen for digestive enzymes analysis. The collected intestines were rinsed in ice-cold Phosphate-Buffered Saline (PBS) (pH 7.5; 1 g per 10 mL), homogenized and centrifuged at 8000 rpm for 5 min (Long et al., 2015). Then the collected supernatant was kept at 4 °C until the preparation of solutions for the analysis of digestive enzyme activity.

2.4. Digestive enzyme activity

The protease activity was evaluated using Folin phenol reagent, and amylase activity was measured using the iodine solution to detect non-hydrolyzed starch, according to Jiang (1982) and Worthington (1993). Based on the protocol described by Borlongan (1990) and Jin (1995) with olive oil as a substratum, the specific activity of lipase was assessed.

2.5. Ammonia acute challenge

After the feeding trial, ten fish with similar size from each aquarium was kept for acute ammonia challenge. A stock tank was filled with water and mixed with ammonium chloride (NH_4Cl) and kept overnight for dichlorination. The amount of ammonium chloride was calculated to ensure the final concentration of total ammonia nitrogen (TA-N) at 5 mg TA-N/L (0.18 mg $\text{NH}_3\text{-N/L}$) depending on water pH and temperature (Abdel-Tawwab et al., 2020; Benli et al., 2008; Hegazi and Hasanein, 2010). Fish were kept under acute ammonia challenge for 6 h. Then, serum and liver samples were collected following the same procedure mentioned above for further analysis.

2.6. Blood biochemical, immunological, and antioxidative analysis

Blood glucose and cortisol levels (MG/100 mL) were determined using enzymatic PAP kits obtained from Bio-Merieux (France) (Trinder, 1969).

Leukocyte phagocytic function was analyzed by following the method of Cai et al. (2004). The number of leukocytes that engulfed bacteria was counted as percentages in relation to the total leukocyte number in the smear from the phagocytosis assay.

The lysozyme activity of sera was assayed according to the method described by Demers and Bayne (1997). The unit of lysozyme present in serum ($\mu\text{g/mL}$) was obtained by matching with the standard curve made with lyophilized hen egg-white lysozyme.

Superoxide dismutase (SOD), catalase (CAT), glutathione peroxidase (GPx), and malonaldehyde (MDA) levels in serum were measured using diagnostic reagent kits following the manufacturer's (Cusabio Biotech Co., Ltd.; China) instructions.

2.7. Gene expression

Total RNA was extracted from 50 mg of the liver using Trizol (iNtRON Biotechnology, Inc., Korea) according to the manufacturer's manual. For this, the total RNA of the samples was first extracted using Trizol (iNtRON Biotechnology). After confirmation of the obtained RNA quality and quantity by Nanodrop (Uv-Vis spectrophotometer Q5000/Quawell, USA). Afterwards, Complementary DNA (cDNA) was synthesized using SensiFAST™ cDNA synthesis kit (Bioline, United Kingdom) according to the manufacturer's protocol.

Gene-specific primer sequences were used for heat shock protein 70 (HSP70), interleukin 1 β (IL-1 β), interleukin 8 (IL-8), and tumor necrosis factor (TNF- α), while β -actin used as a housekeeping gene (Dawood et al., 2020d) (Table 2). The quantitative real-time PCR (Stratagene MX3000P) was used for gene expression.

SYBR green method was used to quantify the gene expression using RT-PCR (SensiFast SYBR Lo-Rox kit, Bioline). The thermocycling conditions for the reaction were 10 min at 95 °C, followed by 40 cycles of 15 s at 95 °C, 30 min at 60 °C, and finally 5 min at 85 °C for 1 min. For each mRNA, gene expression was corrected by the β -actin content in each sample. After verification of PCR efficiency to be around 100%, the gene expression data were calculated according to the method of Livak and Schmittgen (2001).

2.8. Statistical analysis

The normality and homogeneity of the data were checked (SPSS v22.0). Duncan's multiple range test was used to determine differences among treatments before or after ammonia stress by one-way ANOVA method. Two-way ANOVA model was applied to find out the effects of using MNT, ammonia stress (Am), and their interaction on Nile tilapia

Table 2
Primers used for qRT-PCR analysis.

Gene	Primer sequence 5'-3'	Accession No.
TNF- α	F: 5'-GGAAGCAGCTCCACTCTGATGA-3'	JF957373.1
	R: 5'-CACAGCGTGTCTCCTTCGTTCA-3'	
IL-8	F: 5'-GCACTGCCGCTGCATTAAAG-3'	NM_001279704.1
	R: 5'-GCAGTGGGAGTTGGGAAGAA-3'	
IL-1 β	F: 5'-CAAGGATGACGACAAGCCAACC-3'	XM_003460625.2
	R: 5'-AGCGGACAGACATGAGAGTGC-3'	
HSP70	F: 5'-CATCGCTACGGTCTGGACAA-3'	FJ207463.1
	R: 5'-TGCGTCTTCAATGGTCAGGAT-3'	
β -actin	F: 5'-CCACACAGTGCCTCATCTACGA-3'	EU887951.1
	R: 5'-CCACGCTCTGTGAGGATCTTCA-3'	

TNF- α = tumor necrosis factor alpha, IL-8 = interleukin 8, IL-1 β = interleukin 1 beta, HSP70 = heat shock protein 70, β -actin = internal reference gene (house-keeping gene).

performances. Polynomial contrasts were used to detect linear and quadratic effects of MNT. Using the final body weight, FCR, lysozyme activity, and SOD, the MNT level was calculated by polynomial regression analysis (Yossa and Verdegem, 2015).

3. Results

3.1. Growth performance and digestive enzyme activity

Final body weight (FBW), weight gain (WG), and specific growth rate (SGR) of Nile tilapia fed MNT for eight weeks displayed significantly higher values than fish fed MNT free diet ($P < 0.05$) (Table 3). The highest FBW, WG, and SGR have observed in fish fed MNT at the rate of 0.20% and 0.30%. Conversely, the feed conversion ratio (FCR) was significantly reduced in fish fed MNT at the rate of 0.20% when compared to the control without significant differences with the other groups ($P > 0.05$). The survival rate had no significant differences among the groups ($P > 0.05$).

The regression analysis revealed that the optimum rate of MNT required for the best FBW is 0.26% (Fig. 1A). The optimum dietary supplementation of MNT needed for the lowest FCR value is 0.30% (Fig. 1B).

The protease activity had a higher value in fish fed MNT at the rate of 0.30% than the control group ($P < 0.05$) without differences with fish fed the remaining levels of MNT (Table 4). Dietary MNT did not influence the amylase and lipase activities ($P > 0.05$).

3.2. Glucose and cortisol levels

Before the ammonia challenge, the glucose level was significantly reduced in fish fed MNT at the rate of 0.20%, 0.30%, and 0.40% without significant difference with fish fed 0.10% MNT ($P > 0.05$) (Table 5). In the same line, cortisol level was significantly reduced in fish fed MNT at the rate of 0.20%, and 0.30% without significant difference with fish fed 0.10% or 0.40% of MNT ($P > 0.05$) (Table 5).

After ammonia challenge, the glucose level was increased in fish fed MNT at the rate of 0 or 0.40% when compared to the same levels before the challenge. Fish fed MNT displayed lower glucose levels than fish fed MNT free diet regardless of the rate of supplementation (Table 5). Additionally, fish exposed to ammonia challenge showed higher cortisol values than fish before the challenge, and the lowest glucose level was observed in fish fed MNT at the rate of 0.30% ($P < 0.05$) (Table 5).

Two-way ANOVA showed that MNT, ammonia stress, and their interaction had a significant effect on the glucose and cortisol levels ($P < 0.05$) (Table 5).

3.3. Blood immunity

The lysozyme (LZM), phagocytic activities (PA), and phagocytic index (PI) were significantly reduced after the ammonia challenge ($P < 0.05$) (Table 6). Before ammonia challenge, fish fed MNT at the rate of 0.20% or 0.30% displayed higher LZM and PI than fish fed MNT free diet whereas fish fed the rate of 0.30% exhibited the highest PA ($P < 0.05$) (Table 6).

After ammonia challenge, tilapia fed MNT at varied levels displayed higher LZM, and PI than the control with the highest being in fish supplied with MNT at the rate of 0.20% or 0.30% ($P < 0.05$) (Table 6). The PA was also significantly enhanced in fish fed MNT at the rate of 0.10% or 0.20% after ammonia challenge ($P < 0.05$) (Table 6).

The regression analysis revealed that the optimum rate of MNT required for the best LZM is 0.22% (Fig. 1C). Two-way ANOVA showed that MNT, ammonia stress, and their interaction had a significant effect on the PA, PI, and LZM ($P < 0.05$) (Table 6).

Table 3
Growth performance, nutrient utilization and survival of Nile tilapia fed varied levels of MNT for 8 weeks.

Item	MNT (%)					P value	
	0%	0.10%	0.20%	0.30%	0.40%	Linear	Quadratic
IBW (g)	15.08 ± 0.04	15.18 ± 0.09	15.13 ± 0.04	15.15 ± 0.05	15.12 ± 0.04	0.212	0.351
FBW (g)	48.77 ± 0.14 ^c	50.69 ± 0.58 ^b	53.11 ± 0.68 ^a	52.47 ± 0.29 ^a	51.47 ± 0.28 ^{ab}	0.001	0.001
WG (%)	223.32 ± 1.50 ^c	233.87 ± 4.09 ^b	250.97 ± 4.81 ^a	246.32 ± 2.35 ^a	240.47 ± 2.45 ^{ab}	0.011	0.021
SGR (%/day)	1.96 ± 0.01 ^c	2.01 ± 0.02 ^b	2.09 ± 0.02 ^a	2.07 ± 0.01 ^a	2.04 ± 0.01 ^{ab}	0.013	0.025
FCR	1.61 ± 0.03 ^a	1.55 ± 0.09 ^{ab}	1.39 ± 0.02 ^b	1.42 ± 0.01 ^{ab}	1.44 ± 0.03 ^{ab}	0.024	0.028
Survival (%)	95.56 ± 2.22	100.00 ± 0.00	97.78 ± 2.22	100.00 ± 0.00	97.78 ± 2.22	0.062	0.071

* Different superscripts refer to differences between all groups for each parameter ($P < 0.05$). Weight gain (WG, %) = $100 \times [\text{final body weight (FBW, g)} - \text{initial body weight (IBW, g)}] / \text{initial body weight (g)}$; Specific growth rate (SGR, %/day) = $100 \times [\ln \text{FBW (g)} - \ln \text{IBW (g)}] / \text{days}$; Feed conversion ratio (FCR) = total dry feed intake (FI, g) / (FBW – IBW); Survival (%) = $100 \times \text{final fish number} / \text{initial fish number}$.

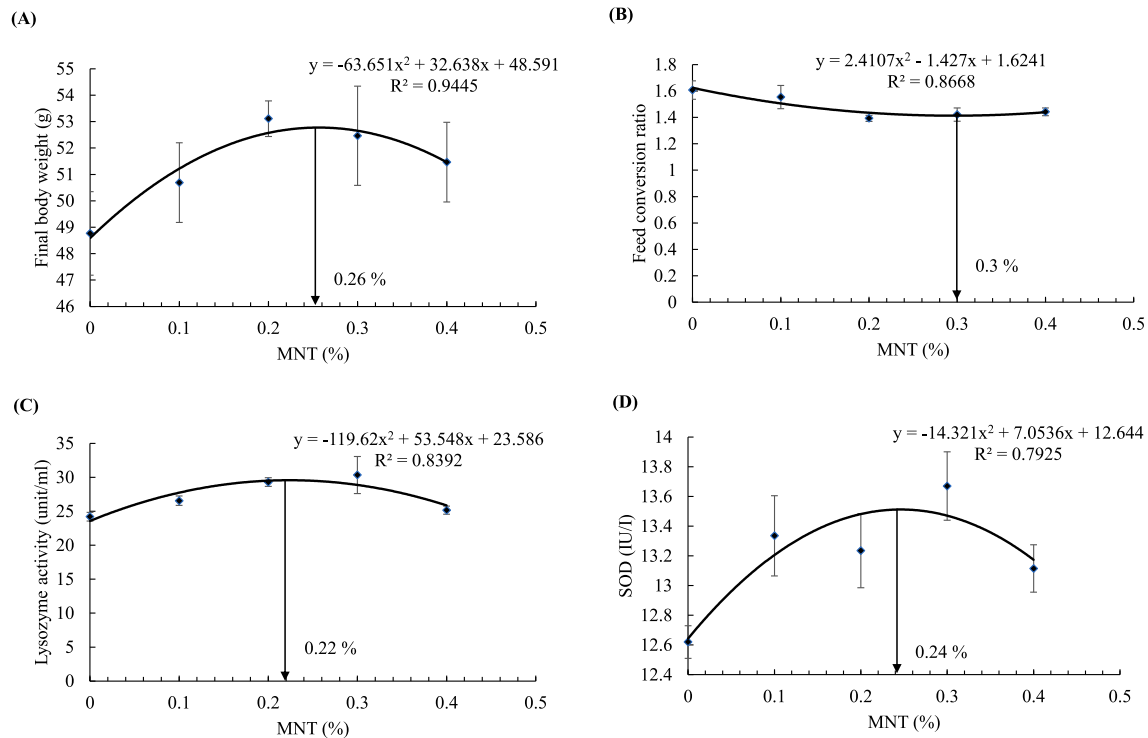


Fig. 1. Significant quadratic relationships and polynomial regressions analysis ($P < 0.05$) between final body weight (A), feed conversion ratio (B), lysozyme activity (C), and superoxide dismutase (SOD) (D) of Nile tilapia and dietary levels of MNT. Values expressed as means ± SE.

3.4. Antioxidative status

Overall, the measured blood SOD, CAT, GPx, and MDA were significantly increased in fish after ammonia challenge when compared to fish before ammonia challenge ($P < 0.05$) (Table 7). Before the ammonia challenge, SOD was significantly increased in fish fed MNT at varying levels, and the CAT has enhanced in fish fed MNT at the rate of 0.30% while the GPx increased in fish fed MNT at 0.20% and 0.30% ($P < 0.05$) (Table 7). Conversely, the MDA level was decreased in fish

fed MNT at the rate of 0.20% and 0.30% ($P < 0.05$) (Table 7). After ammonia challenge, the SOD activity was enhanced in fish fed MNT at the rate of 0.20% and 0.30% while the rate of 0.30% activated CAT and GPx. The MDA was reduced in fish fed MNT with the lowest being in fish fed the level of 0.30% ($P < 0.05$) (Table 7).

The regression analysis revealed that the optimum rate of MNT required for the best SOD is 0.24% (Fig. 1D). Two-way ANOVA showed that MNT, ammonia stress, and their interaction had a significant effect on the SOD, CAT, GPx, and MDA ($P < 0.05$) (Table 7).

Table 4
Digestive enzymes activity in Nile tilapia fed varied levels of MNT for 8 weeks.

Item	MNT (%)					P value	
	0%	0.10%	0.20%	0.30%	0.40%	Linear	Quadratic
Protease activity (unit/mg)	6.18 ± 0.14 ^b	6.75 ± 0.21 ^{ab}	7.36 ± 0.61 ^{ab}	8.63 ± 0.62 ^a	7.24 ± 0.42 ^{ab}	0.012	0.035
Amylase activity (unit/mg)	5.27 ± 0.11	5.62 ± 0.33	6.25 ± 0.32	6.47 ± 1.01	6.22 ± 0.21	0.113	0.231
Lipase activity (unit/mg)	6.12 ± 0.63	5.63 ± 0.21	6.33 ± 0.63	7.05 ± 0.78	6.33 ± 0.21	0.112	0.231

* Values expressed as means ± SE ($n = 3$). Different superscript letters indicate significant differences for each pairwise comparison between treatments.

Table 5

Blood glucose and cortisol levels of Nile tilapia fed varied levels of MNT for 8 weeks and subjected to ammonia stress.

Ammonia stress (Am)	MNT	Glucose (mg/dl)	Cortisol (ng/dl)
Before Am	0%	12.07 ± 0.47 ^{Ba}	26.92 ± 0.65 ^{Ba}
	0.10%	11.35 ± 0.18 ^{Bab}	24.09 ± 0.45 ^{Bab}
	0.20%	10.95 ± 0.36 ^{Bb}	23.89 ± 0.19 ^{Bb}
	0.30%	10.17 ± 0.23 ^{Bb}	23.10 ± 0.27 ^{Bb}
	0.40%	11.02 ± 0.12 ^{Bb}	24.42 ± 0.36 ^{Bab}
P value	Linear	0.001	0.001
	Quadratic	0.001	0.001
After Am	0%	16.66 ± 0.58 ^{Aa}	32.61 ± 0.29 ^{Aa}
	0.10%	13.46 ± 0.44 ^{Ab}	30.91 ± 0.38 ^{Aab}
	0.20%	13.34 ± 0.49 ^{Ab}	30.61 ± 0.35 ^{Aab}
	0.30%	13.02 ± 0.29 ^{Ab}	27.96 ± 0.24 ^{Ab}
	0.40%	14.11 ± 0.29 ^{Ab}	30.20 ± 0.01 ^{Aab}
P value	Linear	0.001	0.001
	Quadratic	0.01	0.001
Two-way ANOVA (P value)			
MNT		0.001	0.001
Am		0.001	0.001
MNT × Am		0.001	0.001

* Capital letters refer to differences between all groups for each parameter, while small superscript letters indicate significant differences for fish fed varied levels of MNT before or after Am ($P < 0.05$). Two-way ANOVA (P value) was calculated if significances are existed.

3.5. Gene expression

The expression of *IL-8* has significantly decreased in fish fed MNT at the rate of 0.30% and 0.40% before the ammonia challenge ($P < 0.05$). Fish fed MNT at varied levels displayed downregulated *IL-8* after ammonia challenge with the lowest being in fish fed at the rate of 0.30% (Fig. 2A). Expression of *IL-8* increased in fish fed MNT at 0%, 0.30%, or 0.40% and exposed to ammonia when compared to the same fish without ammonia stress.

The expression of *TNF-α* has significantly decreased in fish fed MNT at the rate of 0.20% and 0.30% before the ammonia challenge ($P < 0.05$). Fish fed MNT at varied levels displayed downregulated *IL-8* after ammonia challenge (Fig. 2B). Expression of *IL-1β* has significantly lowered in fish fed MNT at the rate of 0.10%, 0.20%, and 0.30% before and after ammonia exposure ($P < 0.05$) (Fig. 2C). The expression of *HSP70* has significantly decreased in fish fed MNT at the rate of 0.20%, 0.30, and 0.40% before the ammonia challenge. Fish fed MNT at varied levels displayed down-regulated *HSP70* after ammonia

challenge (Fig. 2D). The expression of *TNF-α*, *IL-1β*, and *HSP70* significantly increased in fish after ammonia exposure compared with fish before the exposure ($P < 0.05$).

Two-way ANOVA showed that MNT, ammonia stress, and their interaction had a significant effect on the *TNF-α*, *IL-1β*, and *IL-8* ($P < 0.05$) (Fig. 2). However, no interaction effects between dietary MNT and ammonia stress were observed on the expression of *HSP70* gene ($P > 0.05$).

4. Discussion

Aquatic animals reared under highly intensive conditions suffering from several stressors (Dawood et al., 2020b; Dawood et al., 2020c), including the accumulation of ammonia in the rearing water (Abdel-Tawwab et al., 2020). The toxicity with ammonia would impair the immune system and induce inflammation in aquatic animals (Cheng et al., 2015). From a nutritional point of view, using functional feed additives might reduce the impacts of ammonia exposure on aquatic animals (Baba et al., 2018; Dawood et al., 2019b; Yousefi et al., 2020; Zemheri-Navruz et al., 2020). In the present study, MNT was included in tilapia diets reared under optimal conditions for eight weeks; then, fish was exposed to acute ammonia challenge. The obtained results displayed enhanced growth performance and feed efficiency in the case of fish fed MNT at the ration of 0.2% to 0.3%. Further, the regression analysis revealed that the level of 0.26% is recommended for achieving the highest final body weight, while the level of 0.3% is recommended for the lowest FCR value. Concurrent with the present study, Dawood et al. (2020e) reported enhanced growth performance of Nile tilapia fed MNT at the rate of 0.25%. Also, Caspian white fish (*Rutilus frisii kutum*) fed diets with menthol extracts displayed enhanced growth performance (Adel et al., 2015). Conversely, Hoseini et al. (2019) reported that common carp fed menthol for 30 days did not show increased growth performance. The differences are probably due to the experimental conditions, fish species, and feeding duration. The improved growth performance is associated with MNT essential oil's role in increasing the palatability and the attractability of feeds due to the presence of menthol fragrance (Abdel-Latif et al., 2020; Zhang et al., 2020). Additionally, herbal essential oils are known by their capacity to suppress the activity of pathogenic microorganisms in the gastrointestinal tract (GIT) behind their influence on the activity of the digestive enzymes (Cao et al., 2020; Dhama et al., 2018). In this sense, the feed utilization inside the GIT can be enhanced, and consumed feed (FI) would be increased. The reduced FCR value in tilapia fed MNT refers to

Table 6

Blood immune responses of Nile tilapia fed varied levels of MNT for 8 weeks and subjected to ammonia stress.

Ammonia stress (Am)	MNT	Lysozyme activity (unit/ml)	Phagocytic activity (%)	Phagocytic index
Before Am	0%	24.21 ± 0.64 ^{Ab}	22.53 ± 0.55 ^{Ab}	2.02 ± 0.06 ^{Ab}
	0.10%	26.56 ± 0.69 ^{Aab}	23.82 ± 0.26 ^{Aab}	2.28 ± 0.03 ^{Aab}
	0.20%	29.30 ± 0.64 ^{Aa}	24.29 ± 0.65 ^{Aab}	2.57 ± 0.02 ^{Aa}
	0.30%	30.35 ± 2.72 ^{Aa}	25.65 ± 0.35 ^{Aa}	2.54 ± 0.12 ^{Aa}
	0.40%	25.17 ± 0.59 ^{Aab}	24.31 ± 0.02 ^{Aab}	2.07 ± 0.07 ^{Ab}
P value	Linear	0.001	0.001	0.001
	Quadratic	0.001	0.001	0.001
After Am	0%	13.78 ± 0.70 ^{Bc}	15.26 ± 0.57 ^{Bb}	1.55 ± 0.05 ^{Bc}
	0.10%	18.83 ± 0.08 ^{Bb}	16.07 ± 0.10 ^{Ba}	1.73 ± 0.03 ^{Bb}
	0.20%	22.61 ± 0.55 ^{Ba}	17.23 ± 0.38 ^{Ba}	1.90 ± 0.01 ^{Ba}
	0.30%	21.99 ± 0.35 ^{Ba}	15.51 ± 0.08 ^{Bb}	1.96 ± 0.06 ^{Ba}
	0.40%	18.27 ± 0.17 ^{Bb}	15.41 ± 0.49 ^{Bb}	1.72 ± 0.02 ^{Bb}
P value	Linear	0.001	0.001	0.001
	Quadratic	0.001	0.001	0.001
Two-way ANOVA (P value)				
MNT		0.001	0.001	0.001
Am		0.001	0.001	0.001
MNT × Am		0.001	0.001	0.001

* Capital letters refer to differences between all groups for each parameter, while small superscript letters indicate significant differences for fish fed varied levels of MNT before or after Am ($P < 0.05$). Two-way ANOVA (P value) was calculated if significances are existed.

Table 7Antioxidative responses of Nile tilapia fed varied levels of MNT for 8 weeks and subjected to ammonia stress^a.

Ammonia stress (Am)	MNT	SOD (IU/I)	CAT (IU/I)	GPx (IU/I)	MDA (IU/I)
Before Am	0%	12.62 ± 0.16 ^{Bb}	21.23 ± 0.65 ^{Bb}	19.25 ± 0.58 ^{Bb}	20.92 ± 0.48 ^{Ba}
	0.10%	13.34 ± 0.13 ^{Ba}	23.04 ± 0.49 ^{Bab}	21.96 ± 0.42 ^{Bab}	18.84 ± 0.26 ^{Bab}
	0.20%	13.24 ± 0.01 ^{Ba}	23.41 ± 0.09 ^{Bab}	23.19 ± 0.47 ^{Ba}	17.89 ± 0.51 ^{Bb}
	0.30%	13.67 ± 0.13 ^{Ba}	24.84 ± 0.36 ^{Ba}	24.02 ± 0.07 ^{Ba}	17.11 ± 0.95 ^{Bb}
	0.40%	13.12 ± 0.08 ^{Ba}	23.25 ± 0.57 ^{Bab}	22.66 ± 0.70 ^{Bab}	19.16 ± 0.65 ^{Bab}
	<i>P</i> value	0.001	0.001	0.001	0.001
After Am	0%	15.74 ± 0.29 ^{Ab}	24.96 ± 0.23 ^{Ab}	23.65 ± 0.77 ^{Ab}	25.10 ± 0.31 ^{Aa}
	0.10%	15.27 ± 0.55 ^{Ab}	25.67 ± 0.32 ^{Ab}	25.77 ± 0.26 ^{Ab}	21.97 ± 1.01 ^{Ab}
	0.20%	17.41 ± 0.47 ^{Aa}	26.90 ± 0.38 ^{Aab}	25.29 ± 0.62 ^{Aab}	21.46 ± 0.08 ^{Ab}
	0.30%	18.12 ± 0.14 ^{Aa}	27.40 ± 0.67 ^{Aa}	27.12 ± 0.45 ^{Aa}	19.05 ± 0.47 ^{Ac}
	0.40%	15.28 ± 0.57 ^{Ab}	26.45 ± 0.45 ^{Aab}	24.86 ± 0.29 ^{Aab}	22.72 ± 0.23 ^{Ab}
	<i>P</i> value	0.001	0.001	0.001	0.001
Two-way ANOVA (<i>P</i> value)					
MNT		0.001	0.001	0.001	0.001
Am		0.001	0.001	0.001	0.001
MNT × Am		0.001	0.001	0.001	0.001

^a Capital letters refer to differences between all groups for each parameter, while small superscript letters indicate significant differences for fish fed varied levels of MNT before or after Am ($P < 0.05$). Two-way ANOVA (*P* value) was calculated if significances are existed.

the increased WG in tilapia by optimum utilization of FI. In this regard, the activity of protease was enhanced in fish fed MNT at the rate of 0.30%, which correlated with the enhanced growth rate and FCR.

Before and after Nile tilapia were exposed to ammonia stress, the glucose and cortisol levels were lower in fish fed MNT than fish fed the MNT free diet. The glucose and cortisol levels are usually increased in the blood to help the organism counteract with stressful conditions (Barton, 2000; Dawood et al., 2017). Under the circumstances of the current feeding trial, it seems that MNT is required to relieve the impacts of ammonia toxicity. However, toxicity with ammonia increased the glucose and cortisol levels in tilapia compared with the values before the stress. Similarly, Yousefi et al. (2020) reported increased glucose and cortisol levels in common carp exposed to ammonia stress. It could be explained by MNT's role in reducing the metabolism of the energy required to cope with the impacts of ammonia exposure in tilapia. In this regard, Taheri Mirghaied et al. (2019) reported that common carp fed diets with 1,8-cineole essential oil and exposed to ammonia challenge displayed reduced glucose and cortisol levels in the blood.

Lysozyme (LZM) and phagocytosis (PI) are immune responses that required to counteract with the infection. The current study revealed higher LZM and PI in fish fed MNT than fish fed MNT free diet, which indicates increased immune responses of fish. In agreement with the current study, Caspian white fish fed menthol extract displayed enhanced LZM (Adel et al., 2015). After the ammonia challenge, tilapia showed reduced LZM and PI than before the stress test. The suppressed immune system is one of the features of ammonia toxicity (Abdel-Tawwab et al., 2020), which resulted from increased oxidative stress and the level of lipid peroxidation in the fish's body (Yousefi et al., 2020). However, after ammonia challenge, fish fed MNT displayed higher LZM and PI than fish fed MNT free diet. This result can be attributed to MNT's role in activating the immune response and scavenging the excessive reactive oxygen metabolites (ROS). Similarly, MNT reduced the impact of chlorpyrifos toxicity on the LZM and PA in Nile tilapia (Dawood et al., 2020e).

Under stressful conditions, including ammonia toxicity, fish are suffering from oxidative stress, which associated with the influence of stress on the balance between production and removal of ROS from the cell (Pamplona and Costantini, 2011). The interruption in the scavenging ability induces lipid peroxidation, accumulation of ROS, and breaking the body cells' lipid membranes (Burgos-Aceves et al., 2018a). This interruption also includes the damage of the DNA of immune cells (Burgos-Aceves et al., 2018b). In this case, the body secretes enzymatic

and non-enzymatic antioxidative responses (SOD, CAT, and GPx) to diminish the ROS and enhance the antioxidative capacity (dos Reis Goes et al., 2019). However, the failure to do that causes lipid peroxidation and malondialdehyde production (MDA) (Silvestre, 2020). In the present study, the SOD, CAT, and GPx activities were increased, but the MDA level decreased in tilapia fed MNT essential oil before and after ammonia challenge. Further, the SOD, CAT, and GPx activities and MDA levels were increased in tilapia after ammonia challenge compared to fish before the challenge. Concurrent with the present study, Hoseini et al. (2019) reported that common carp fed diets with menthol inhibited the adverse effects of ammonia toxicity on the antioxidative responses (SOD, CAT, GPx, and MDA). In the same line, Dawood et al. (2020e) reported that tilapia fed MNT and exposed with chlorpyrifos exhibited upregulation of antioxidative related genes. The antioxidative capacity of MNT is associated with the high content of polyphenols and flavonoids, which acts as a natural antioxidant that redox excessive ROS (Suttili et al., 2018).

Inflammatory (*IL-8*, *TNF-α*, and *IL-1β*) and stress (*HSP70*) related genes in the present study displayed upregulated mRNA levels in tilapia after ammonia exposure, but feeding with MNT essential oil succeeded to balance and reduce the (*IL-8*, *TNF-α*, *IL-1β*, and *HSP70*) either before or after ammonia stress. Cytokines related genes inhibit the inflammation features by enhancing immunity (Jia et al., 2019), while *HSP70* is upregulated during stressful conditions to reduce the adverse impacts of stress on the fish body (Jun et al., 2015). The downregulated *IL-8*, *TNF-α*, *IL-1β*, and *HSP70* genes in fish fed MNT indicates the balance between the inflammatory and anti-inflammatory responses attributed to the potent role of MNT as a natural immunostimulant (Dawood et al., 2020e). The present study displayed exclusive and exciting results about MNT essential oil's potential impact to cope with the adverse effects of ammonia exposure through the regulation of *IL-8*, *TNF-α*, *IL-1β*, and *HSP70* genes in the liver of Nile tilapia.

5. Conclusion

To sum up, the present study investigated the potential role of MNT essential oil in improving the growth performance and protease activity of Nile tilapia. Additionally, the continuous feeding of MNT for eight weeks resulted in regulating the pro-inflammatory (*IL-8*, *TNF-α*, and *IL-1β*) and stress (*HSP70*) related genes, blood cortisol and glucose levels, lysozyme and phagocytic activities, and antioxidative responses (SOD, CAT, GPx, and MDA) to overcome with the adverse impacts of ammonia toxicity. The recommended dose of MNT essential oil is 0.22% to 0.30%

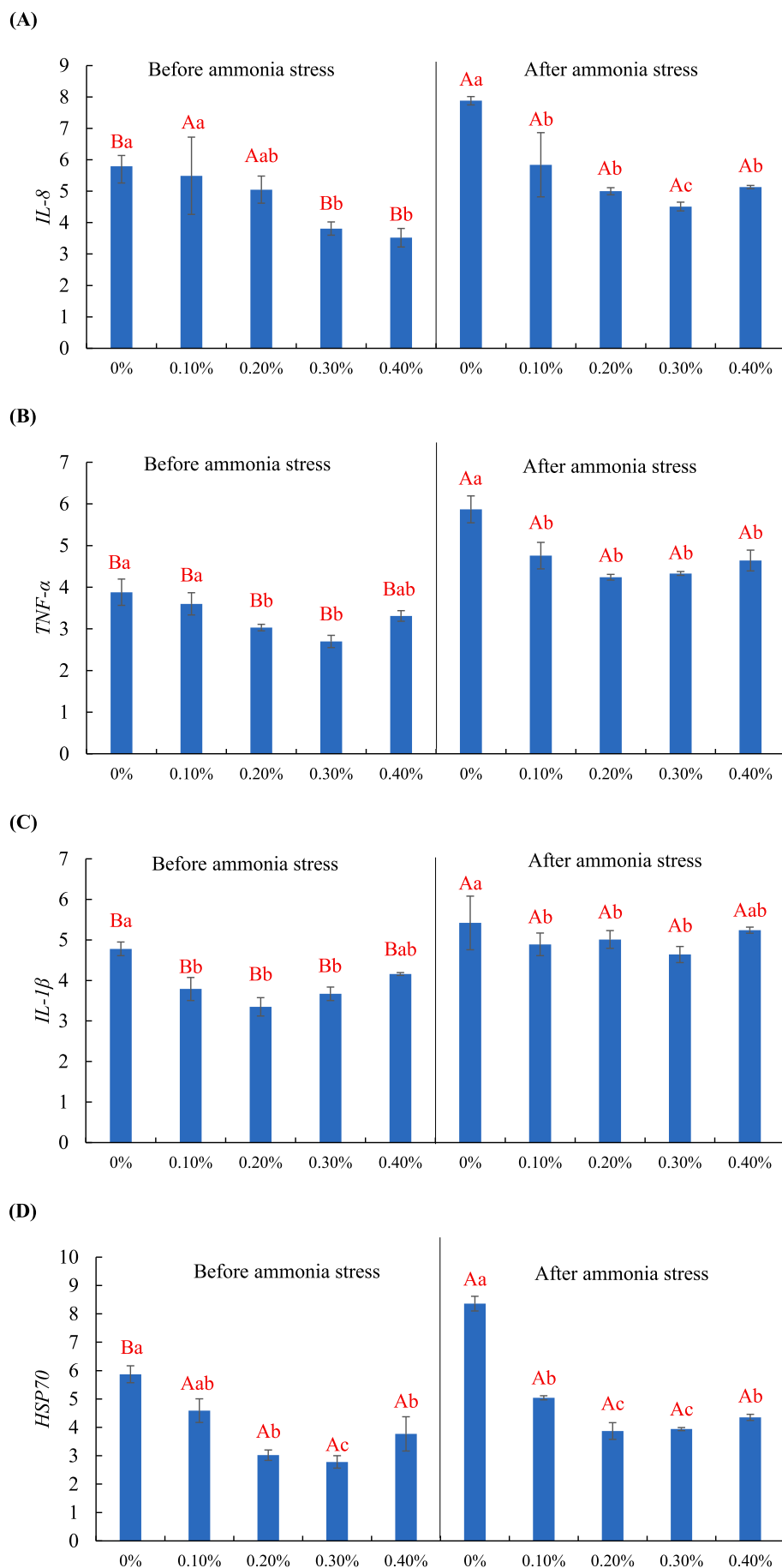


Fig. 2. Relative transcription of (A) *interleukin 8* (*IL-8*), (B) *tumor necrosis factor alpha* (*TNF-α*), (C) *interleukin 1 beta* (*IL-1β*), and (D) *heat shock protein 70* (*HSP70*) of Nile tilapia fed varied levels of MNT for 8 weeks and subjected to ammonia stress. Bars with capital letters refer to differences between all groups for each parameter, while small letters indicate significant differences for fish varied levels of MNT before or after ammonia stress (Am) (as mean $\pm$ SE, $n = 3$) (P value linearly and quadratically < 0.05). Two-way ANOVA (P value) for *TNF-α* (MNT:0.001; Am:0.006; MNT $\times$ Am: 0.007), *INF-γ* (MNT:0.001; Am:0.001; MNT $\times$ Am: 0.005), *IL-1β* (MNT:0.001; Am:0.008; MNT $\times$ Am: 0.002), and *HSP70* (MNT:0.0001; Am:0.002; MNT $\times$ Am: 0.708).

in the diets of Nile tilapia.

Declaration of Competing Interest

The authors declare no competing interests.

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Author statement

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Declaration of Competing Interest

The authors declare no conflicts of interest.

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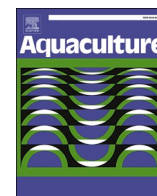
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Update

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The authors regret expressing the gross energy in Table 1 as kcal/kg while it should be as MJ/kg.

The authors would like to apologise for any inconvenience caused.

Table 1. Basal diet and proximate chemical composition (on dry matter basis).

Ingredient	(%)	Chemical composition	(%)
Fish meal (65%)	10	Crude protein	30.14
Soybean meal (44%)	36	Crude lipids	6.98
Gluten	5	Ash	7.57
Wheat bran	12	Crude fibers	5.45

(continued on next column)

(continued)

Ingredient	(%)	Chemical composition	(%)
Rice bran	10	Gross energy (MJ/kg) ^b	18.44
Yellow corn	12		
Wheat flour	8.92		
Fish oil	3		
Vitamin and mineral mix ^a	2		
Dicalcium phosphate	1		
Vitamin C	0.08		

^a Vitamin and mineral mixture detailed by Dawood et al. (2020b).

^b Gross energy was calculated based on the values of values for protein, lipid, and carbohydrate as 23.6, 39.5 and 17.2 KJ/g, respectively.

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